

The Role of Improving Directions in Mixed Integer Bilevel Linear Optimization

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The Setting

- *First-level (aka Leader) variables*: $x \in X \subseteq \mathbb{Z}^{r_1} \times \mathbb{R}^{n_1-r_1}$
- *Second-level (aka Follower) variables*: $y \in Y \subseteq \mathbb{Z}^{r_2} \times \mathbb{R}^{n_2-r_2}$

Mixed Integer Bilevel Linear Optimization Problem

$$\min_{x,y} \left\{ cx + d^1 y \mid x \in X, y \in \mathcal{P}_1(x), y \in \operatorname{argmin} \{ d^2 z \mid z \in \mathcal{P}_2(x) \cap Y \} \right\}, \quad (\text{MIBLP})$$

where

$$\begin{aligned} \mathcal{P}_1(x) &= \left\{ y \in \mathbb{R}_+^{n_2} \mid G^1 y \geq b^1 - A^1 x \right\}, \\ \mathcal{P}_2(x) &= \left\{ y \in \mathbb{R}_+^{n_2} \mid G^2 y \geq b^2 - A^2 x \right\}. \end{aligned}$$

Assumptions

- 1 All input data are integer
- 2 All first-level variables are integer and appear in second-level constraints
- 3 The feasible regions are all bounded

The Notation

Let us denote

Rational Reaction Set

$$\mathcal{R}(x) = \left\{ y \in \mathcal{S}(x) \mid d^2 y \leq d^2 \bar{y}, \forall \bar{y} \in \mathcal{S}(x) \right\}$$

$\Rightarrow$ **follower** will respond **optimally** to **leader's** decision.

where

- $\mathcal{P} = \{(x, y) \in \mathbb{R}^{n_1} \times \mathbb{R}^{n_2} \mid y \in \mathcal{P}_1(x) \cap \mathcal{P}_2(x)\},$
- $\mathcal{S} = \{(x, y) \in X \times Y \mid (x, y) \in \mathcal{P}\},$
- $\mathcal{S}(x) = \{y \in Y \mid (x, y) \in \mathcal{S}\} \Rightarrow$ **follower's** feasible points for a given **leader's** decision.

Bilevel Feasible Points

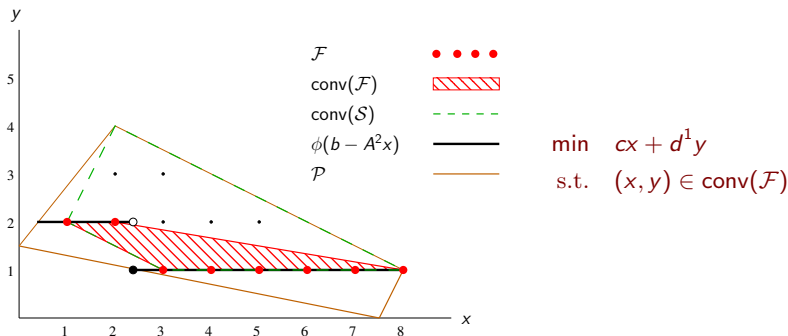
$$\mathcal{F} = \{(x, y) \in \mathbb{R}^{n_1} \times \mathbb{R}^{n_2} \mid x \in X, y \in \mathcal{R}(x)\}$$

Optimistic Assumption

Whenever $|\mathcal{R}(x)| > 1$, **follower** selects the response most favorable for the **leader**.

Theorem 1 (Tahernejad and Ralphs [2020]). *Under the stated assumptions, we have that*

$$\min_{(x,y) \in \mathcal{F}} cx + d^1 y = \min_{(x,y) \in \text{conv}(\mathcal{F})} cx + d^1 y. \quad (1)$$



- Reformulation (1) suggests a Branch-and-Cut algorithm, similar to that for MILPs [DeNegre and Ralphs, 2009]
- Dual bounds can be obtained by optimizing over a relaxed feasible region
- The goal is to approximate $\text{conv}(\mathcal{F})$ with linear inequalities

Components

- **Bounding**
 - **Dual bound** $\Rightarrow$ A “tractable” relaxation strengthened with valid inequalities
 - **Primal bound** $\Rightarrow$ **Feasible solutions**
- Branching $\Rightarrow$ Valid disjunctions
- **Cut generation** $\Rightarrow$ Valid inequalities for $\text{conv}(\mathcal{F})$
- Search strategies
- Preprocessing methods
- Primal heuristics
- Control mechanisms

Challenges

- Both checking feasibility of integer solutions and separation using “simple” valid inequalities are (related) NP-complete problems.
- The complexity of both can be seen as arising from the core problem of identifying *second-level improving (feasible) directions*.
- Separately checking feasibility of the solution to the relaxation and then generating violated valid inequalities if infeasible is time-consuming.

Second-level Improving Directions

Let $(\hat{x}, \hat{y}) \in \mathcal{P}$. For $w \in \mathbb{Z}^{r_2} \times \mathbb{R}^{n_2-r_2}$, we say that w is

- an *improving direction* (ID) if $d^2 w < 0$, and
- a *feasible direction* (FD) w.r.t. $(\hat{x}, \hat{y})$ if $\hat{y} + w \in \mathcal{P}_2(\hat{x})$.

The set of all IDs is

$$\mathcal{W} = \left\{ w \in \mathbb{Z}^{r_2} \times \mathbb{R}^{n_2-r_2} \mid d^2 w < 0 \right\}.$$

The set of all *improving feasible directions* (IFDs) w.r.t. $(\hat{x}, \hat{y})$ is

$$\mathcal{W}(\hat{x}, \hat{y}) = \{ w \in \mathcal{W} \mid \hat{y} + w \in \mathcal{P}_2(\hat{x}) \}.$$

Bilevel Feasibility Oracle

Lemma 1. For $(\hat{x}, \hat{y}) \in \mathcal{S}$, we have $(\hat{x}, \hat{y}) \in \mathcal{F} \iff \mathcal{W}(\hat{x}, \hat{y}) = \emptyset$.

- The standard feasibility check is to determine whether an improving solution exists.
- By checking whether $\mathcal{W}(\hat{x}, \hat{y}) = \emptyset$ instead, we gain flexibility and increase efficiency.
 - We can generate cuts cheaply using the resulting IFD, unifying two important subproblems.
 - By restricting the search for IFDs, we obtain a variety of relaxations, which gives us a way to tune the difficulty of the oracle problem.

Improving Directions (Example)

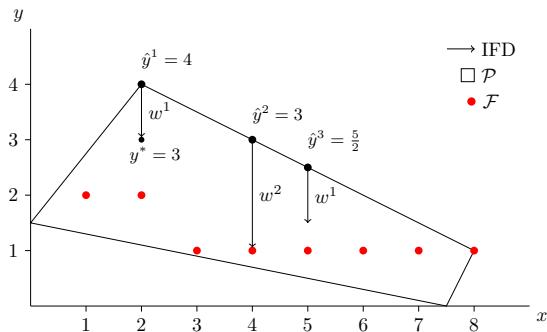


Figure: The improving feasible directions moving points in S (or P) towards points in F .

k-opt Hierarchy

For $Y = \{0, 1\}^{n_2}$, Shi et al. [2023] proposed the restriction known as the *k-opt neighborhood*

$$\mathcal{N}_k(y) = \{\bar{y} \in \{0, 1\}^{n_2} \mid \|\bar{y} - y\|_1 \leq k\}, \text{ for } k \in \mathbb{Z}_+, k \leq n_2,$$

from which we can derive the *k-optimal reaction set*

$$\mathcal{R}_k(x) = \left\{ y \in \mathcal{S}(x) \mid d^2 y \leq d^2 \bar{y}, \forall \bar{y} \in \mathcal{N}_k(y) \cap \mathcal{S}(x) \right\}$$

and the *k-opt feasible set*

$$\mathcal{F}(k) = \{(x, y) \in \mathbb{R}^{n_1} \times \mathbb{R}^{n_2} \mid x \in X, y \in \mathcal{R}_k(x)\}.$$

The following theorem shows that the hierarchy is finite.

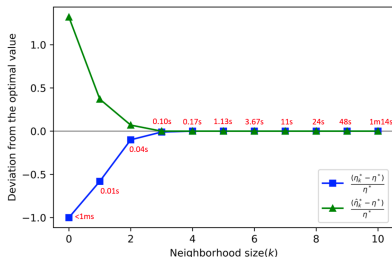
Finiteness

Theorem 2. *There exists a $\bar{k} \in \mathbb{N}$ such that*
 $\mathcal{S} = \mathcal{F}(0) \supseteq \mathcal{F}(1) \supseteq \mathcal{F}(2) \supseteq \dots \supseteq \mathcal{F}(\bar{k}) = \mathcal{F}.$

k -opt Relaxation

$$\begin{aligned}
 \min \quad & cx + d^1 y \\
 \text{s.t.} \quad & (x, y) \in X \\
 & y \in \mathcal{R}_k(x)
 \end{aligned} \tag{k-BP}$$

Let us look at an example of Knapsack Interdiction Problem from Shi et al. [2023]

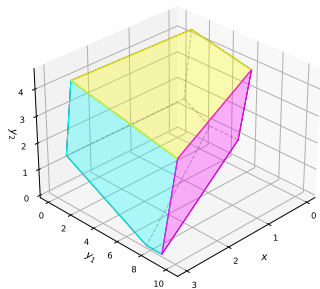


At a glance

- (k-BP) can be formulated as a MILP;
- Dual bounds are tighter than $\mathcal{S}$ and $\mathcal{P}$, even for small k ;
- (k-BP) converges to (MIBLP) rather fast;
- (k-BP) bound comes at a reasonable computational cost.

Interpreting the Hierarchy

- By Theorem 2, the hierarchy partitions the points in $\mathcal{F}$ into sets of points on a finite number of different “levels.”
- The hierarchy can be associated in a natural way with a hierarchy of valid inequalities, as we will see.

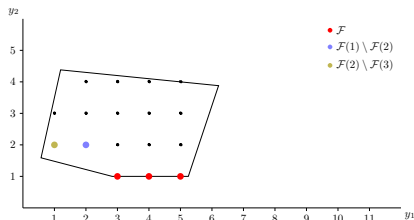


$$\begin{aligned} \min_{x \in \{0,1,2,3\}} \quad & -x - 2y_1 - 5y_2 \\ \text{s.t.} \quad & y \in \operatorname{argmin} \{y_2 : \end{aligned}$$

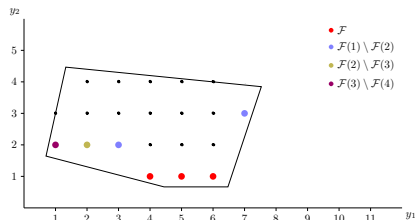
$$\begin{aligned} & x + 3y_2 \geq 3 \\ & -16x + 12y_1 - 4y_2 \leq 59 \\ & -x + 9y_1 - 2y_2 \geq 2 \\ & -2x + 6y_1 + 23y_2 \geq 40 \\ & -x + y_1 + 10y_2 \leq 45 \\ & y \in \mathbb{Z}_+^2 \} \end{aligned}$$

Figure: A three-dimensional MIBLP and the feasible region of its LP relaxation.

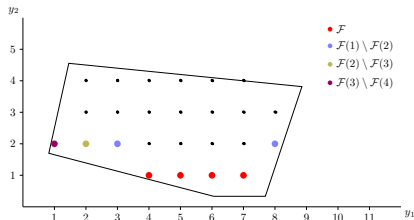
Visualizing the Hierarchy



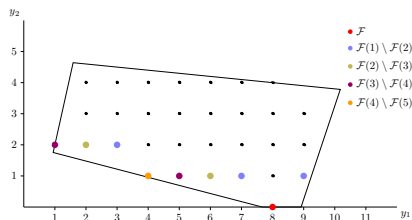
(a) $\hat{x} = 0$



(b) $\hat{x} = 1$



(c) $\hat{x} = 2$



(d) $\hat{x} = 3$

Figure: Slices of the feasible region of the example.

Valid Inequality

The triple $(\alpha^x, \alpha^y, \beta) \in \mathbb{R}^{n_1+n_2+1}$ is a *valid inequality* for $\mathcal{F}$ if

$$\mathcal{F} \subseteq \{(x, y) \in \mathbb{R}^{n_1+n_2} \mid \alpha^x x + \alpha^y y \geq \beta\}.$$

We refer to a valid inequality for $\mathcal{F}$ that is violated by a given solution of the current relaxation as a *cutting plane* (cut).

Bilevel Free Set

A *bilevel free set* (BFS) is a set $C \subseteq \mathbb{R}^{n_1+n_2}$ such that $\text{int}(C) \cap \mathcal{F} = \emptyset$.

General Recipe for Valid Inequalities

- Find a BFS $C \subseteq \mathbb{R}^{n_1+n_2}$;
- Then inequalities valid for $\text{conv}(\overline{\text{int}(C)})$ are also valid for $\mathcal{F}$.

Improving Direction Intersection Cuts (IDICs)

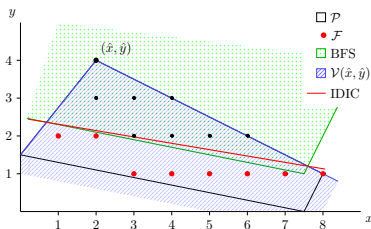
Bilevel Free Set [Fischetti et al., 2018]

Let $(\hat{x}, \hat{y})$ be an extreme point of $\mathcal{P}$ and let $w \in \mathcal{W}(\hat{x}, \hat{y})$ ($\Leftarrow$ IFD). Then

$$C(w) = \left\{ (x, y) \in \mathbb{R}^{n_1+n_2} \mid A^2x + G^2y \geq b^2 - G^2w - 1, y + w \geq -1 \right\}.$$

is a BFS $\Rightarrow$ the set of points (x, y) such that w is an IFD w.r.t. (x, y) (plus a small buffer)

- We consider the radial cone $\mathcal{V}(\hat{x}, \hat{y})$ with vertex $(\hat{x}, \hat{y})$ containing $\mathcal{P}$ arising from the optimal basis to the LP relaxation.



- $\alpha^x x + \alpha^y y = \beta$ is the hyperplane passing through the intersection of $C(w)$ and $\mathcal{V}(\hat{x}, \hat{y})$.
- $(\alpha^x, \alpha^y, \beta)$ is valid for $\text{conv}(\mathcal{F})$.
- $\|w\|_2$ affects the “depth” of the cut
- separation is not guaranteed when $(\hat{x}, \hat{y}) \in \mathcal{P} \setminus \mathcal{S}$.

Note that the BFS shown above is **without** the buffer.

k -IDICs

Denoting the set of all improving directions with a 1-norm no bigger than k as

$$\mathcal{W}^k = \{w \in \mathcal{W} \mid \|w\|_1 \leq k\},$$

we have the following definition.

Definition 1. For $k \in [\bar{k}]$, an $\mathcal{F}(k)$ -free set is a BFS $C(w)$ such that $w \in \mathcal{W}^k$.

Definition 2. For $k \in [\bar{k}]$, a k -IDIC is an IC generated from an $\mathcal{F}(k)$ -free set.

k -IDICs and the k -opt Feasible Set

Theorem 3. For all $k \in [\bar{k}]$, the following statements hold:

- (1) all k -IDICs are valid for $\mathcal{F}(k)$; and
- (2) for all $(x, y) \in \mathcal{S} \setminus \mathcal{F}(k)$, there exists a k -IDIC violated by (x, y) .

In other words, k -IDICs can separate all and only points in $\mathcal{S} \setminus \mathcal{F}(k)$.

- Note that this does not say that k -IDICs describe $\text{conv}(\mathcal{F}(k))$.
- Rather, it says roughly that the set of all k -IDICs is a valid formulation for (k-BP).
- From these concepts it is possible to define notions of rank and closures that mirror the familiar ones from mixed integer linear optimization.

- We can describe $\mathcal{W}(\hat{x}, \hat{y})$ as the points satisfying

$$\begin{aligned}d^2 w &\leq -1 \\ G^2 w &\geq b^2 - A^2 \hat{x} - G^2 \hat{y} \\ w &\geq -\hat{y} \\ w &\in \mathbb{Z}^{r_2} \times \mathbb{R}^{n_2 - r_2}.\end{aligned}\tag{IFD}$$

- (IFD) is formally equivalent to solving the **follower's** problem.
- However, we can plug (IFD) with a variety of obj. functions to obtain directions with favorable properties, e.g.,
 - **min 1** $\Rightarrow$ Checking Bilevel Feasibility
 - **min $d^2 w$** $\Rightarrow$ Find the best **follower's** solution for the given $\hat{x}$
- Or, we can solve (IFD) with some kind of local search and optimize nonlinear obj. functions such as
 - **min $\|w\|_2$** $\Rightarrow$ Find the “shortest” IFD.
- Gaar et al. [2024] highlights the benefits of heuristic methods for finding IFDs.

- IDs unify the **bilevel feasibility check** with the **generation** of strong inequalities.
- We can solve MIBLPs without ever solving the follower's problem explicitly.
- For fixed k , (IFD) is polynomially solvable.
- Solving to (integer) optimality while only generating k -IDICs is equivalent to solving the k -opt relaxation.
- This gives us an adjustable way of generating feasible solutions with quality guarantees.
- There is much more left to explore.

MibS [Tahernejad et al., 2020]

Branch&Cut open-source solver for MIBLPs (available at github.com/coin-or/MibS)

- **Bounding**
 - **Dual bound** $\Rightarrow$ Compute optimal $(\hat{x}, \hat{y}) \in \mathcal{P}$ (LP relaxation)
 - **Primal bound** $\Rightarrow$ **Feasible solutions** by solving follower's problem (a MILP)
- **Cut generation**
 - **IDICs** $\Rightarrow$ find w by solving (IFD) exactly or with local search
 - MILP cuts $\Rightarrow$ when $(\hat{x}, \hat{y}) \in \mathcal{P} \setminus \mathcal{S}$
- Feasibility check is done by solving second-level problem.

idB&C

Implemented modifying MibS to unify the oracles.

- **Bounding**
 - **Dual bound** $\Rightarrow$ Compute optimal $(\hat{x}, \hat{y}) \in \mathcal{P}$ (LP relaxation)
 - **Primal bound** $\Rightarrow$ **Feasible solutions** when $(\hat{x}, \hat{y}) \in \mathcal{S}$ and $\mathcal{W}(\hat{x}, \hat{y}) = \emptyset$
- **Cut generation**
 - **IDICs** $\Rightarrow$ find w by solving (IFD) exactly or with local search
 - MILP cuts $\Rightarrow$ when $(\hat{x}, \hat{y}) \in \mathcal{P} \setminus \mathcal{S}$
- Feasibility check is done by using result of solving (IFD).

Configurations

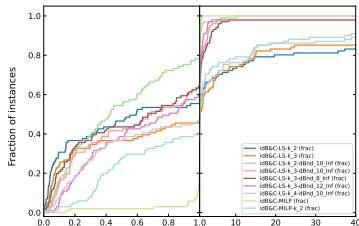
- Mibs: version 1.2 default ($\Leftarrow$ the “baseline”)
- MibS w/IDICs: version 1.2 with only IDICs turned on.
- idB&C: MibS w/ unified oracle.

Dataset

- The BENCHMARK is composed of 277 pure-integer instances from BOBILib (available at bobilib.org)
- Instances that required a Solution Time $\in]0, 1] \cup [3600, \infty[$ seconds for all configurations are excluded from the plots

Solution Time vs MibS w/IDICs

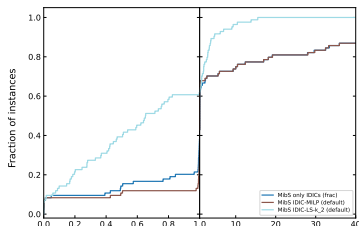
Baseline Profile: CPU Time (F-D-D2-Z-ZZ)



Ratio of baseline: MibS only IDICs (frac)

Solution Time vs MibS default

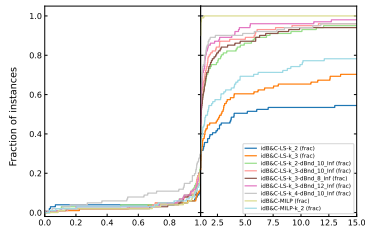
Baseline Profile: CPU Time (F-D-D2-Z-ZZ)



Ratio of baseline: MibS (default)

Nodes Processed vs MibS w/IDICs

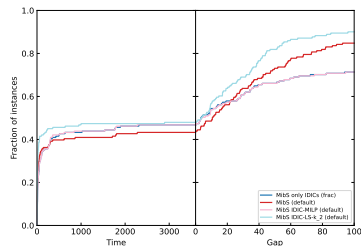
Baseline Profile: Nodes Processed (F-D-D2-Z-ZZ)



Ratio of baseline: MibS only IDICs (frac)

Time-Gap Cumulative Profile

Cumulative Profile: Time-Gap (F-D-D2-Z-ZZ)



Final Remarks

- Simply replacing the standard oracle for feasibility with the oracle for IFDs does not harm performance.
- Our B&C based on a unified oracle based on IFDs shows promising results.
- In particular, when the search for IFDs is combined with a Local Search it achieves both lower **Cut Generation** and **Solution Time**.
- There is a tradeoff between the effort spent generating IFDs and the strength of the resulting IDICs.
- This tradeoff is related to the k -opt hierarchy.

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Thanks for your attention!

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